Matched asymptotic expansions for shallow turbulent flows: from Richard, Couderc & Vila (2023) to permeable, rough beds

Derivation notes

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Abstract

Part 1 re-derives the depth-averaged model of Richard, Couderc & Vila (2023, *J. Fluid Mech.* 954, A24; hereafter RCV) by matched asymptotic expansions. Parts 2–5 redo the derivation over a permeable bed described by the double-averaged equations of Rousseau & Ancey (2022, *J. Fluid Mech.* 941, A51; hereafter RA): first with a null Reynolds stress in the bed, then with the Li & Sawamoto mixing length, then with the dispersive stress. The main result is structural: *if the bed and its roughness layer are thin compared with the depth, all of the bed physics is confined to the inner layer and enters the depth-averaged model only through two constants of an inner boundary-value problem*, R_p and R_{1p}, which replace RCV's R and R_1. Everything in the outer layer, the matching logic and the structure of the final system (mass, momentum, enstrophy) is unchanged. What changes from step to step is the inner problem and therefore $R_p(d^+)$ and $R_{1p}(d^+)$, which I compute numerically. Every intermediate result of RCV that is quoted here was re-derived symbolically (1-D) or numerically and agrees with the paper; two points where I found something worth flagging are marked in grey boxes.

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1 Richard, Couderc & Vila (2023): the smooth-bed derivation

A note on the request. RCV treat the *smooth* turbulent case; there is no roughness in that paper. Roughness enters here only through the permeable-bed parts (the roughness layer of RA), where it turns out to be handled by the same machinery.

1.1 Governing equations and closure

A turbulent flow runs down a plane inclined at θ; Oz is normal to the bed and $\boldsymbol{g} = g(\sin\theta\cos\beta, \sin\theta\sin\beta, -\cos\theta)$. Reynolds-averaged mass and momentum:

$$\operatorname{div}\boldsymbol{v} = 0, \qquad \rho\Big(\partial_t\boldsymbol{v} + \operatorname{div}(\boldsymbol{v}\otimes\boldsymbol{v})\Big) = \rho\boldsymbol{g} - \mathbf{grad}\,p + \operatorname{div}\boldsymbol{\tau}, \qquad \boldsymbol{\tau} = 2\rho(\nu+\nu_T)\boldsymbol{D}, \tag{1}$$

with Prandtl's mixing length, Van Driest damping and the Nezu–Rodi free-surface reduction (wake neglected):

$$\nu_T = \sqrt{2}\,L_m^2\sqrt{\boldsymbol{D}:\boldsymbol{D}}, \qquad L_m = \kappa z\big(1-e^{-z^+/A^+}\big)\sqrt{1-z/h}, \qquad z^+ = zu_b/\nu,\ u_b = \sqrt{\tau_b/\rho}. \tag{2}$$

Boundary conditions: $\boldsymbol{v} = 0$ at $z = 0$; kinematic condition $w(h) = \partial_t h + u(h)\partial_x h + v(h)\partial_y h$; stress-free surface.

1.2 Scalings

With $\varepsilon = h_N/L \ll 1$, horizontal velocities scaled by u_N, w by εu_N, p by $\rho g h_N$, shear stresses by $\rho\kappa^2u_N^2$, normal stresses by $\varepsilon\rho\kappa^2u_N^2$, ν_T by $\kappa^2h_Nu_N$, L_m by κh_N, define

$$\eta = \frac{\nu}{\kappa^2h_Nu_N} = \varepsilon^{2+m}\ (m>0), \qquad \lambda = \frac{\sin\theta}{\kappa^2F^2}, \qquad \boldsymbol{\lambda} = \lambda(\cos\beta,\sin\beta), \qquad s = z/h. \tag{3}$$

Outer (shallow-water) scaling. Dropping primes, $\nu_{\text{eff}} = \eta + \nu_T$ and, because $z^+ = z\sqrt{\tau_b}/(\eta\kappa)$ is huge, the Van Driest factor is 1:

$$\nu_{\text{eff}} = z^2(1-s)\,\|\partial_z\boldsymbol{u}\| + O(\varepsilon^2), \qquad \partial_z\boldsymbol{\tau}_{sh} = -\boldsymbol{\lambda} + \frac{\varepsilon}{\kappa^2}\Big[\partial_t\boldsymbol{u} + \operatorname{div}(\boldsymbol{u}\otimes\boldsymbol{u}) + \partial_z(w\boldsymbol{u}) + \frac{1}{F^2}\,\mathbf{grad}\,p\Big] + O(\varepsilon^2), \tag{4}$$

$\partial_z p = -\cos\theta + O(\varepsilon)$, $p(h) = O(\varepsilon)$, $\boldsymbol{\tau}_{sh}(h) = O(\varepsilon^2)$, with $\boldsymbol{u} = (u,v)$ and $\boldsymbol{\tau}_{sh} = (\tau_{xz},\tau_{yz})$. The no-slip condition cannot be imposed in this scaling.

Inner (viscous) scaling. $\tilde z = z/\eta$, $\tilde w = w/\eta$, other variables unchanged. Then $\tilde\nu_{\text{eff}} = 1 + \tilde\nu_T$ and

$$\partial_{\tilde z}\tilde{\boldsymbol{\tau}}_{sh} = O(\eta), \qquad \partial_{\tilde z}\tilde p = O(\eta), \qquad \tilde{\boldsymbol{\tau}}_{sh} = \Big[1 + \tilde z^2(1-s)\big(1-e^{-\xi/A}\big)^2\|\partial_{\tilde z}\tilde{\boldsymbol{u}}\|\Big]\partial_{\tilde z}\tilde{\boldsymbol{u}}, \tag{5}$$

with $\xi = 2\sqrt{\tilde\tau_b}\,\tilde z = 2\kappa z^+$ and $A = 2\kappa A^+$.

Every variable is expanded as $X = X_0 + \varepsilon X_1 + O(\varepsilon^2)$.

1.3 Order 0

Outer. Integrating (4) with $\boldsymbol{\tau}_{sh}(h) = 0$:

$$\boldsymbol{\tau}_{sh}^{(0)} = \boldsymbol{\lambda}h(1-s). \tag{6}$$

The constitutive law gives $z^2(1-s)\|\partial_z\boldsymbol{u}_0\|\partial_z\boldsymbol{u}_0 = \boldsymbol{\lambda}h(1-s)$, i.e. $\|\partial_z\boldsymbol{u}_0\| = \sqrt{\lambda h}/z$: the free-surface factor $(1-s)$ of L_m cancels the linear stress decrease exactly, which is why the outer profile is a pure logarithm,

$$\boldsymbol{u}_0 = \boldsymbol{u}_0(h) + \sqrt{\lambda h}\,\hat{\boldsymbol{\lambda}}\ln s. \tag{7}$$

Inner. $\tilde{\boldsymbol{\tau}}_{sh}^{(0)}$ is constant, equal to its outer limit $\boldsymbol{\lambda}h$; so $\tilde\tau_b = \lambda h$. Solving the quadratic constitutive law for $\|\partial_{\tilde z}\tilde{\boldsymbol{u}}_0\|$ (secant form, regular at $\xi = 0$):

$$\frac{\partial\tilde{\boldsymbol{u}}_0}{\partial\tilde z} = \frac{2\boldsymbol{\lambda}h}{1+\sqrt{\Delta}}, \quad \Delta = 1 + \xi^2(1-e^{-\xi/A})^2 \implies \tilde{\boldsymbol{u}}_0 = \sqrt{\lambda h}\,\hat{\boldsymbol{\lambda}}\,G_0(\xi), \quad G_0(\xi) = \int_0^\xi \frac{\mathrm{d}\xi'}{1+\sqrt{\Delta}}. \tag{8}$$

Since $\int_0^\xi \mathrm{d}\xi'/(1+\sqrt{1+\xi'^2}) = \ln(\xi+\sqrt{1+\xi^2}) - \xi/(1+\sqrt{1+\xi^2})$,

$$G_0(\xi) = -\frac{\xi}{1+\sqrt{1+\xi^2}} + \ln\left(\xi+\sqrt{1+\xi^2}\right) + R(\xi) \;\xrightarrow[\xi\to\infty]{}\; \ln 2\xi - 1 + R,$$
$$R = \int_0^\infty \Big[\frac{1}{1+\sqrt{\Delta}} - \frac{1}{1+\sqrt{1+\xi^2}}\Big]\mathrm{d}\xi. \tag{9}$$

Near the wall $G_0 \simeq \xi/2$, i.e. $u^+ = z^+$.

Matching. Overlap at $s = b\sqrt{\eta}$, i.e. $\xi = 2b\sqrt{\lambda h^3}/\sqrt{\eta}$. Equating $\boldsymbol{u}_0(s = b\sqrt{\eta})$ and $\tilde{\boldsymbol{u}}_0(\xi\to\infty)$, the $\ln b$ cancels and

$$\boldsymbol{u}_0(h) = \sqrt{\lambda h}\,\hat{\boldsymbol{\lambda}}\left(R - 1 + \ln 2 + \ln M - \ln\eta\right), \qquad M = 2\sqrt{\lambda h^3}, \tag{10}$$

with matching error $O(\sqrt{\eta}) = o(\varepsilon)$ because $m > 0$.

Second small parameter. $\mu = -2/\ln\eta = 2/\ln(\kappa^2 Re)$, with $\varepsilon \ll \mu^p$ for all p. Taking $\boldsymbol{u}_0 = O(1)$ requires $\lambda = \mu^2\lambda_0$, $\lambda_0 = O(1)$ (hence $\sin\theta = O(\mu^2)$). Then

$$\boldsymbol{u}_0 = \sqrt{\lambda_0 h}\,\hat{\boldsymbol{\lambda}}\left[2 + \mu(Q + \ln s)\right], \qquad Q \equiv R - 1 + \ln 2 + \ln M, \tag{11}$$

$$\boldsymbol{U}_0 = \langle\boldsymbol{u}_0\rangle = \sqrt{\lambda_0 h}\,\hat{\boldsymbol{\lambda}}\,C(\mu), \qquad C(\mu) = 2 + \mu(R - 2 + \ln 2 + \ln M). \tag{12}$$

In dimensional 1-D form, $u_0/u_b = \kappa^{-1}\ln z^+ + B$ with

$$\kappa B = R - 1 + 2\ln 2 + \ln\kappa. \tag{13}$$

Check. For $\kappa = 0.41$, $A^+ = 26$ I obtain $R = 2.6689$, $B = 5.277$ (RCV: 2.67, 5.28).

The vertical velocity w_0 follows from continuity and the kinematic condition (RCV eq. 3.32), and $p_0 = (h - z)\cos\theta$, $\tilde{p}_0 = h\cos\theta$.

1.4 Order 1: outer layer

At $O(\varepsilon)$, (4) reads $\kappa^2\partial_z\boldsymbol{\tau}_{sh}^{(1)} = \partial_t\boldsymbol{u}_0 + \operatorname{div}(\boldsymbol{u}_0\otimes\boldsymbol{u}_0) + \partial_z(w_0\boldsymbol{u}_0) + F^{-2}\,\mathbf{grad}\,p_0$. Integrating from h (where $\boldsymbol{\tau}_{sh}^{(1)} = 0$) and using the kinematic condition to absorb the Leibniz boundary terms,

$$\kappa^2\boldsymbol{\tau}_{sh}^{(1)} = \partial_t\int_h^z \boldsymbol{u}_0\,\mathrm{d}z + \operatorname{div}\int_h^z \boldsymbol{u}_0\otimes\boldsymbol{u}_0\,\mathrm{d}z + w_0\boldsymbol{u}_0 + \frac{1}{F^2}\,\mathbf{grad}\int_h^z p_0\,\mathrm{d}z, \tag{14}$$

with $\partial_t h = -\operatorname{div}(h\boldsymbol{U}_0)$. Inserting (11) gives

$$\boldsymbol{\tau}_{sh}^{(1)} = \frac{\lambda_0 h}{\kappa^2}\hat{\boldsymbol{\lambda}}(\hat{\boldsymbol{\lambda}}\cdot\mathbf{grad}\,h)\left[1 - s + \mu T_1(s) + \mu^2 T_2(s)\right] + \frac{h}{\kappa^2}\frac{\cos\theta}{F^2}\,\mathbf{grad}\,h\,(s-1). \tag{15}$$

I re-derived T_1, T_2 symbolically for a *general* $Q(h)$ (this generality is needed in Part 2). With

$$q_1 \equiv h\,\frac{\mathrm{d}Q}{\mathrm{d}h}, \qquad P \equiv \frac{Q}{2} + q_1, \tag{16}$$

the result is remarkably compact:

$$T_1(s) = (1-s)(2P-1) + \tfrac{1}{2}s\ln s, \qquad T_2(s) = (1-s)\big(P^2 - P - \tfrac{7}{4}\big) + s\ln s\big(\tfrac{P}{2} - \tfrac{7}{4} + \tfrac{1}{2}\ln s\big). \tag{17}$$

For the smooth bed $q_1 = 3/2$ (from $\ln M \propto \frac{3}{2}\ln h$), so $2P = R + 2 + \ln 2 + \ln M$, and (17) reproduces RCV (A4)–(A5) exactly.

Linearised constitutive law. Write $\boldsymbol{\tau} = \mathcal{F}(\|\partial_z\boldsymbol{u}\|)\,\partial_z\boldsymbol{u}/\|\partial_z\boldsymbol{u}\|$. Perturbing about $\boldsymbol{u}_0$, the component of $\partial_z\boldsymbol{u}_1$ parallel to $\hat{\boldsymbol{\lambda}}$ is governed by the *tangent* compliance $1/\mathcal{F}'$ and the normal component by the *secant* compliance $\|\partial_z\boldsymbol{u}_0\|/\mathcal{F}$. In the outer layer $\mathcal{F} = z^2(1-s)\|\partial_z\boldsymbol{u}\|^2$, so $\mathcal{F}' = 2\,\mathcal{F}/\|\partial_z\boldsymbol{u}\|$, and with $\|\partial_z\boldsymbol{u}_0\| = \sqrt{\lambda h}/z$:

$$\frac{\partial\boldsymbol{u}_1}{\partial z} = \frac{1}{2\kappa^2}\sqrt{\frac{\lambda_0}{h}}\,\hat{\boldsymbol{\lambda}}(\hat{\boldsymbol{\lambda}}\cdot\mathbf{grad}\,h)\frac{1 - s + \mu T_1 + \mu^2 T_2}{\mu s(1-s)} + \frac{1}{2\kappa^2}\sqrt{\frac{\lambda_0}{h}}\frac{\cos\theta}{\lambda_0 F^2}\frac{\hat{\boldsymbol{\lambda}}(\hat{\boldsymbol{\lambda}}\cdot\mathbf{grad}\,h)}{\mu s} - \frac{1}{\kappa^2}\sqrt{\frac{\lambda_0}{h}}\frac{\cos\theta}{\lambda_0 F^2}\frac{\mathbf{grad}\,h}{\mu s}, \tag{18}$$

(RCV A14). Its integral $\boldsymbol{u}_1 = \boldsymbol{u}_1(h) + \int_1^s(\cdot)\,\mathrm{d}s$ contains $\mathrm{Li}_2(1-s)$, $\mathrm{Li}_2(s)$, $\mathrm{Li}_3(s)$ and $\zeta(3)$ (RCV A15) and diverges like $\ln s/\mu$ as $s\to 0$.

1.5 Order 1: inner layer and matching

In the inner layer $\partial_{\tilde z}\tilde{\boldsymbol\tau}_{sh}^{(1)}=0$, so $\tilde{\boldsymbol\tau}_{sh}^{(1)}=\boldsymbol\tau_{sh}^{(1)}(0)$: *the inner order-1 problem is the linear response of the inner solution to a perturbation of the wall stress.* Split $\boldsymbol\tau_{sh}^{(1)}(0)$ into its part along $\hat{\boldsymbol\lambda}$ and its normal part:

- the parallel part is integrated with the tangent compliance $1/\sqrt{\Delta}$: $G_1(\xi)=\displaystyle\int_0^\xi \frac{\mathrm{d}\xi'}{\sqrt{\Delta}} \xrightarrow[\xi\to\infty]{} \ln 2\xi + R_1$, $\quad R_1=\displaystyle\int_0^\infty\Big[\frac{1}{\sqrt{\Delta}}-\frac{1}{\sqrt{1+\xi^2}}\Big]\mathrm{d}\xi$;
- the normal part is a rigid rotation of the order-0 profile, integrated with the secant compliance $2/(1+\sqrt{\Delta})$, i.e. with G_0.

Hence (RCV A28 rewritten)

$$\tilde{\boldsymbol u}_1=\frac{\hat{\boldsymbol\lambda}\big(\hat{\boldsymbol\lambda}\cdot\boldsymbol\tau_{sh}^{(1)}(0)\big)}{2\mu\sqrt{\lambda_0 h}}\,G_1(\xi)+\frac{\boldsymbol\tau_{sh}^{(1)}(0)-\hat{\boldsymbol\lambda}\big(\hat{\boldsymbol\lambda}\cdot\boldsymbol\tau_{sh}^{(1)}(0)\big)}{\mu\sqrt{\lambda_0 h}}\,G_0(\xi)+O(\mu^2). \tag{19}$$

Matching. At the overlap $\ln s=\ln b-1/\mu$ and $\ln 2\xi=\ln b+\ln(2M)+1/\mu$. Writing the outer solution as $s\to 0$ as $\boldsymbol u_1\sim\boldsymbol u_1(h)+\boldsymbol c\ln s+\boldsymbol c_0$, the coefficient $\boldsymbol c$ coincides automatically with the inner coefficient of $\ln\xi$, and matching the constants gives (1-D, parallel part)

$$u_1(h)=c\Big(\frac{2}{\mu}+\ln 2+\ln M+R_1\Big)-c_0,\qquad c_0=-\frac{a}{\mu}\int_0^1 g\,\mathrm{d}s=\frac{a\pi^2}{12}+O(a\mu), \tag{20}$$

where $a=\sqrt{\lambda_0 h}\,h_x/(2\kappa^2)$ and $g=\mu\ln s/[2(1-s)]+\mu^2\ln s\,(\frac{P}{2}-\frac{7}{4}+\frac{1}{2}\ln s)/(1-s)$ is the integrable part of (18). This is RCV (A31); the normal part brings R instead of R_1, which is why RCV's third line contains R_1-2R.

Flag 1 (convention on the damping function). RCV evaluate the Van Driest factor with the order-0 wall stress, $\tilde\tau_b=\lambda h$, also at order 1. The order-1 inner response then uses the tangent compliance with the damping *frozen*, which yields $R_1=4.82$. If the damping instead follows the corrected wall stress (as eq. (2) literally says), the inner solution is $u=u_b f(z^+)$ and its exact derivative with respect to u_b gives $\partial u/\partial u_b\to\kappa^{-1}[\ln(4\kappa z^+)+R]$: R_1 is replaced by R and $\alpha=R_1-R+1$ becomes 1. Both are legitimate modelling choices (whether turbulence damping adapts instantly to the wall stress); they give the same structure but different relaxation coefficients. Part 2 keeps both options.

1.6 Depth-averaged quantities

Using integration by parts ($\int_0^1 f\,\mathrm{d}s=f(1)-\int_0^1 sf'\mathrm{d}s$) and $\int_0^1 s\ln^k s/(1-s)\,\mathrm{d}s=(-1)^k k!\,[\zeta(k+1)-1]$:

$$\begin{aligned}U_1=a\Big[\frac{2}{\mu^2}+\frac{1}{\mu}\big(R_1-1+\ln 2+\ln M+2T_1(0)\big)-\frac{1}{2}+2T_2(0)+T_1(0)(R_1-1+\ln 2+\ln M)\Big]\\ -a\,\Pi\Big[\frac{2}{\mu^2}+\frac{1}{\mu}(R_1-1+\ln 2+\ln M)\Big]+O(a\mu),\qquad \Pi=\frac{\cos\theta}{\lambda_0 F^2}\quad\text{(1-D form of RCV A33)},\end{aligned} \tag{21}$$

$$\varphi^{(0)}=\frac{1}{h^2}\langle(\boldsymbol u_0-\boldsymbol U_0)\otimes(\boldsymbol u_0-\boldsymbol U_0)\rangle=\frac{\mu^2\lambda_0}{h}\,\hat{\boldsymbol\lambda}\otimes\hat{\boldsymbol\lambda}, \tag{22}$$

$$\varphi^{(1)}=\frac{\lambda_0 h_x}{\kappa^2 h}\big[1-\Pi+\mu(2P-\zeta(3))\big]+O(\mu^2)\qquad\text{(1-D form of RCV 4.11)}. \tag{23}$$

The enstrophy tensor φ is the only new unknown: $\langle\boldsymbol u\otimes\boldsymbol u\rangle=\boldsymbol U\otimes\boldsymbol U+h^2\varphi$, and the cubic moment $\langle\boldsymbol u^*\otimes\boldsymbol u^*\otimes\boldsymbol u^*\rangle=O(\mu^3)$ is dropped (Teshukov's weakly-sheared approximation).

1.7 Momentum equation and inversion

Averaging the momentum equation and using $\boldsymbol\tau_{sh}^{(0)}(0)=h\boldsymbol\lambda$:

$$\partial_t(h\boldsymbol U)+\mathrm{div}(h\boldsymbol U\otimes\boldsymbol U+h^3\varphi)+\mathbf{grad}\Big(\frac{h^2}{2F^2}\cos\theta\Big)=-\kappa^2\boldsymbol\tau_{sh}^{(1)}(0)+O(\varepsilon). \tag{24}$$

The three quantities $\boldsymbol\tau_{sh}^{(1)}(0)$, $\boldsymbol U_1$, $\varphi^{(1)}$ are all linear in $\hat{\boldsymbol\lambda}\cdot\mathbf{grad}\,h$ and $\cos\theta\,\mathbf{grad}\,h/F^2$; eliminating these two gradients expresses the wall stress in terms of $\boldsymbol U_1$ and $\varphi^{(1)}$, which are then replaced by relaxation differences

($\varepsilon \boldsymbol{U}_1 = \boldsymbol{U} - \boldsymbol{U}_0$, etc.). In 1-D I find

$$\tau^{(1)}(0) = c_1 \,\frac{\mu^2}{C^2}\, 2U_0U_1 + c_2\, h^2\varphi^{(1)}, \qquad c_1 = 1 - \frac{\alpha_1}{2}\mu + O(\mu^2), \quad c_2 = -\frac{\alpha_2}{2}\mu + O(\mu^2), \tag{25}$$

$$\alpha = R_1 - R + 1, \qquad \alpha_2 = \frac{1}{2(\zeta(3)-1)}, \qquad \alpha_1 = \alpha - \alpha_2, \tag{26}$$

in agreement with RCV (4.12) at this order ($\alpha = 3.155$, $\alpha_2 = 2.475$, $\alpha_1 = 0.680$ recomputed). The perpendicular parts, which do not appear in 1-D, split c_2 into RCV's $-\alpha\,\mathrm{tr}\,\varphi$ and $+\alpha_1\varphi\cdot\hat{\boldsymbol{\lambda}}$ terms.

Flag 2 (what is asymptotically fixed). The elimination determines c_1 and c_2 unambiguously to $O(\mu)$. RCV write the result in a factorised form, $(1-\alpha_1\mu/C)$ and $-(\alpha/\kappa^2)(\kappa-\alpha_1\mu/C)(\kappa\mu/C)+\alpha_1\mu/C$, which also contains $O(\mu^2)$ pieces. The strict $O(\mu^2)$ coefficients obtained from the same elimination are different and large (they contain $P^2/(\zeta(3)-1)^2$ terms), so I read the $O(\mu^2)$ part of the factorised form as a resummation choice rather than a derived term. Also, in (4.12) the factor printed as $(\kappa-\alpha_1\mu/C)$ is consistent with (4.22) only if read as $\kappa(1-\alpha_1\mu/C)$. Nothing below depends on this: I carry coefficients to the order that is fixed.

With $C_f = \mu^2\kappa^2/C^2(\mu)$ and $\hat{\boldsymbol{g}} = g\sin\theta(\cos\beta, \sin\beta)$, the dimensional system is RCV (4.26)–(4.28):

$$\partial_t h + \mathrm{div}(h\boldsymbol{U}) = 0, \tag{27}$$

$$\begin{aligned}\partial_t(h\boldsymbol{U}) + \mathrm{div}(h\boldsymbol{U}\otimes\boldsymbol{U} + h^3\varphi) + \mathbf{grad}\left(\frac{gh^2}{2}\cos\theta\right) &= \left(1 - \frac{\alpha_1}{\kappa}\sqrt{C_f}\right)(\hat{\boldsymbol{g}}h - C_f\|\boldsymbol{U}\|\boldsymbol{U}) \\ &+\alpha(\kappa - \alpha_1\sqrt{C_f})h\sqrt{C_f}\Big(h\,\mathrm{tr}\varphi - \frac{\hat{g}}{\kappa^2}\Big)\frac{\hat{\boldsymbol{g}}}{\hat{g}} - \kappa\alpha_1 h\sqrt{C_f}\Big(h\varphi\cdot\frac{\hat{\boldsymbol{g}}}{\hat{g}} - \frac{\hat{\boldsymbol{g}}}{\kappa^2}\Big),\end{aligned} \tag{28}$$

plus the enstrophy equation below. The friction law is explicit:

$$\frac{\kappa}{\sqrt{C_f}} = R - 2 + 2\ln 2 + \ln\kappa + \ln\frac{\sqrt{gh^3\sin\theta}}{\nu}. \tag{29}$$

1.8 Enstrophy equation

Forming $\boldsymbol{u}\otimes$(momentum)+(momentum)$\otimes\boldsymbol{u}$, averaging, subtracting $\boldsymbol{U}\otimes$(averaged momentum)$+\ldots$ and dropping $O(\mu^3)$ gives

$$\partial_t(h\varphi) + \mathrm{div}(h\varphi\otimes\boldsymbol{U}) - 2h\varphi\;\mathrm{div}\,\boldsymbol{U} + \mathbf{grad}\,\boldsymbol{U}\cdot h\varphi + h\varphi\cdot(\mathbf{grad}\,\boldsymbol{U})^{\mathrm{T}} = \frac{\kappa^2}{\varepsilon h^2}\big[\boldsymbol{U}\otimes\boldsymbol{\tau}_{sh}(0) + \boldsymbol{\tau}_{sh}(0)\otimes\boldsymbol{U} - 2\boldsymbol{W}\big], \tag{30}$$

with the dissipation tensor $\boldsymbol{W} = \int_0^h \nu_{\mathrm{eff}}\,\partial_z\boldsymbol{u}\otimes\partial_z\boldsymbol{u}\,\mathrm{d}z$. $\boldsymbol{W}$ is not integrated directly: $\boldsymbol{W}_0 = \frac{h}{2}(\boldsymbol{U}_0\otimes\boldsymbol{\lambda} + \boldsymbol{\lambda}\otimes\boldsymbol{U}_0)$, and $\boldsymbol{W}_1$ is *deduced* from the $O(\varepsilon^0)$ part of the averaged second-moment equation (RCV B4) evaluated with order-0 fields. I redid this in 1-D: $2W_1 - 2h\lambda U_1 = e_1\frac{4\mu^2}{C^2}U_0^2U_1 + e_2\,2U_0h^2\varphi^{(1)}$ with $e_1 = 1 - \alpha\mu/2 + O(\mu^2)$ and $e_2 = O(\mu^2)$, consistent with RCV (B11), and the relaxation form (4.28) follows with α_2 and α.

2 Permeable bed: the general framework

2.1 Double-averaged equations (RA)

Let $n(z)$ be the porosity (RA's ϵ; renamed to avoid a clash with the shallowness parameter), equal to n_b in the bed and 1 in the free flow, with a roughness layer of thickness $\sim d_p$ in between. For the intrinsic double-averaged velocity $\boldsymbol{U}$ (RA eq. 2.4) the shear-flow form of the momentum equation is

$$\partial_z\big[\tau_v + \tau_t + \tau_d\big] - \rho\nu\, n'\partial_z U + nf + \rho n\, g\sin\theta + \ldots = 0, \qquad \tau_v = \rho\nu\,\partial_z(nU), \tag{31}$$

where the dots are the unsteady, advective and pressure terms (of the same form as in RCV, weighted by n). The closures used below are

$$\text{drag (Ergun):}\quad nf = -\frac{A_E(1-n)^2}{n\,d_p^2}\rho\nu U - \frac{B_E n(1-n)}{d_p}\rho|U|U, \quad A_E = 180,\ B_E = 1.75, \tag{32}$$

$$\text{turbulent:}\quad \tau_t = \rho n\,\ell_t^2|\partial_z U|\partial_z U, \tag{33}$$

$$\text{dispersive:}\quad \tau_d = \rho n\,\nu_d\,\partial_z U, \quad \nu_d = \lambda_d\, d_p\frac{1-n}{1-n_b}|U|, \quad \lambda_d = 0.015. \tag{34}$$

The Kozeny–Carman permeability is $K = d_p^2 n_b^3/[A_E(1-n_b)^2]$, so $\sqrt{K} \simeq 0.031\, d_p$ for $n_b = 0.4$. The linear drag is written as $n\bar{f} = -\rho\nu n^2 U/K$, i.e. RA's middle form $\bar{f} = -\mu n U/K$, which gives Darcy's law for the superficial velocity, $nU_{\mathrm{SSL}} = gKi/\nu$ (RA 2.9). RA's printed right-hand side of (2.8) has one power of n fewer, so it is not the same expression. The first version of these notes used that printed form. Correcting it changes R_p by at most 0.035, because the Brinkman layer is thin.

2.2 Where the bed sits in the asymptotic structure

Let $\delta_\nu = \nu/u_*$. Define $d^+ = d_p u_*/\nu$ and $Re_K = \sqrt{K}u_*/\nu$ ($\simeq 0.031\, d^+$). Assume

$$\frac{\max(\delta_\nu, d_p)}{h} = O(\eta) \quad \text{(thin roughness layer)}, \qquad d^+,\ Re_K \text{ fixed as } \eta \to 0. \tag{35}$$

In the inner layer (thickness $\delta_i = \max(\delta_\nu, d_p)$), compare each term with $\partial_z \tau \sim \rho u_*^2/\delta_i$:

- gravity, pressure gradient, unsteadiness and advection: relative size $\delta_i/h = O(\eta)$, *negligible* exactly as in RCV;
- Darcy drag over the inner layer: $\nu U/K$ vs u_*^2/δ_ν gives $Re_K^{-2} = O(1)$: *the drag enters the leading-order inner balance*;
- deep-bed Darcy velocity: $U_{\mathrm{SSL}} = gK\sin\theta/\nu$, so $U_{\mathrm{SSL}}/u_* = Re_K^2\, \delta_\nu/h = O(\eta)$;
- subsurface and inner-layer volume flux relative to hU: $O(\eta)$, and $O(\eta\, h_s/h)$ for a bed of thickness h_s.

Consequences: (i) the outer equations, boundary conditions and solutions of Part 1 are *unchanged*; (ii) the depth-averaged mass equation is unchanged to $O(\eta)$; (iii) the outer coordinate may be measured from any level inside the roughness layer, because an $O(d_p)$ shift changes $\ln z$ by $O(d_p/z)$, which vanishes in the overlap; (iv) the inner problem changes, and it is quasi-steady and one-dimensional.

Range of validity. Assumption (35) is *large relative submergence*. RA's experiments and steep mountain streams have $h/d_p = O(1)$, outside the formal regime. The full-depth check of §6 shows that the error made is first order in d_p/h and stays moderate: 5% on the mean velocity at $h/d_p = 3.75$, 1.6% at $h/d_p = 10$. A correction at $O(d_p/h)$ would require the roughness layer in the outer problem (porosity and drag there, depth-averaging weighted by n).

2.3 The order-0 inner problem

In wall units (z^+, $u^+ = U/u_*$, all stresses divided by ρu_*^2), the leading-order inner balance is the steady, gravity-free version of (31):

$$\frac{\mathrm{d}S}{\mathrm{d}z^+} = n'\frac{\mathrm{d}u^+}{\mathrm{d}z^+} + \frac{A_E(1-n)^2}{n\, d^{+2}}u^+ + \frac{B_E n(1-n)}{d^+}u^{+2}, \qquad S = n\Big[1 + \nu_d^+ + \ell_t^{+2}\Big|\frac{\mathrm{d}u^+}{\mathrm{d}z^+}\Big|\Big]\frac{\mathrm{d}u^+}{\mathrm{d}z^+} + n'u^+, \tag{36}$$

with $u^+ \to 0$ in the bulk bed and $S \to 1$ above the roughness layer. Solving the quadratic for the velocity gradient in secant form,

$$\frac{\mathrm{d}u^+}{\mathrm{d}z^+} = \frac{2T}{a + \sqrt{a^2 + 4\ell_t^{+2}|T|}}, \qquad T = \frac{S - n'u^+}{n}, \quad a = 1 + \nu_d^+. \tag{37}$$

Above the roughness layer $n = 1$, $\nu_d = 0$, $S = 1$ and $\ell_t^+ \to \kappa z^+$, so $u^+ \to \kappa^{-1}\ln z^+ + B_p$. **Define**

$$R_p \equiv \kappa B_p + 1 - 2\ln 2 - \ln\kappa = \lim_{z^+\to\infty}\big[\kappa u^+ - \ln(4\kappa z^+) + 1\big], \tag{38}$$

so that $R_p = R$ for RCV's smooth wall. Since $\xi = 2\kappa z^+$ and $\kappa u^+ = G_0$, this is exactly the statement $G_0 \to \ln 2\xi - 1 + R_p$. All geometric parameters scale with d_p, so $R_p = R_p(d^+; n_b, \ldots)$, and I define

$$\gamma \equiv \frac{\mathrm{d}R_p}{\mathrm{d}\ln d^+}. \tag{39}$$

Order 0 then goes through verbatim with $R \to R_p$: $u_0(h)$, (11), (12), $C(\mu) = 2 + \mu(R_p - 2 + \ln 2 + \ln M)$, the log law with B_p, and the explicit friction law

$$\frac{\kappa}{\sqrt{C_f}} = R_p(d^+) - 2 + 2\ln 2 + \ln\kappa + \ln\frac{\sqrt{gh^3\sin\theta}}{\nu}, \qquad d^+ = \frac{d_p\sqrt{gh\sin\theta}}{\nu}. \tag{40}$$

It stays *explicit* because d^+ is built from h and the slope. In the fully rough limit $R_p \simeq c_r - \ln d^+$, and the viscosity cancels: $\kappa/\sqrt{C_f} = c_r - 2 + 2\ln 2 + \ln\kappa + \ln(h/d_p)$.

2.4 The order-1 inner problem

The order-1 inner equations are again the linearisation of (36) about the order-0 state, forced by $\boldsymbol{\tau}_{sh}^{(1)}(0)$; the drag and dispersive terms make it a linear ODE rather than a quadrature, but the decomposition of (19) survives:

- *normal part*: all closures above are isotropic in the horizontal plane (drag $\propto U$ and $|U|U$, $\nu_d \propto |U|$, $\tau_t \propto |\partial_z U|\partial_z U$), so rotating the wall stress rigidly rotates the profile: response G_0, constant R_p;
- *parallel part*: response G_1 of the linearised problem with $\delta S \to 1$ above the roughness layer and $\delta u \to 0$ in the bed, and

$$R_{1p} \equiv \lim_{z^+\to\infty} \Big[2\kappa \frac{\delta u^+}{\delta S} - \ln(4\kappa z^+)\Big]. \tag{41}$$

Two linearisations are possible (Flag 1):

$$\begin{aligned} &\text{consistent:} \quad R_{1p} = R_p + \gamma \quad \Rightarrow \quad \alpha_p = 1 + \gamma; \\ &\text{frozen damping:} \quad R_{1p} \text{ from the ODE with the damping factor held fixed.} \end{aligned} \tag{42}$$

The consistent identity follows from $u = u_* F(z^+; d^+)$: at fixed dimensional z, $\partial u/\partial u_* = F + z^+ F_{z^+} + d^+ F_{d^+} \to \kappa^{-1}[\ln(4\kappa z^+) + R_p + \gamma]$. I checked it numerically: the linearised ODE and $R_p + \gamma$ agree to four decimals.

Order-1 matching is RCV's (20) with $R_1 \to R_{1p}$ for the parallel part and $R \to R_p$ for the normal part.

2.5 What the h-dependence of R_p does to the outer order 1

In RCV, $Q = R - 1 + \ln 2 + \ln M$ depends on h only through $\ln M$. Now $Q = R_p(d^+) - 1 + \ln 2 + \ln M$ and $d^+ \propto \sqrt{h}$, so

$$q_1 = h\frac{\mathrm{d}Q}{\mathrm{d}h} = \frac{3}{2} + \frac{\gamma}{2}, \tag{43}$$

which enters T_1, T_2 through $P = Q/2 + q_1$ in (17), and hence U_1, $\varphi^{(1)}$ and the reconstructed profile. Redoing the elimination (25) symbolically with general (P, q_1, R_p, R_{1p}), I find

$$c_1 = 1 - \frac{R_{1p} - R_p + 1 - \alpha_2}{2}\mu + O(\mu^2), \qquad c_2 = -\frac{\alpha_2}{2}\mu + O(\mu^2), \qquad e_1 = 1 - \frac{R_{1p} - R_p + 1}{2}\mu + O(\mu^2) : \tag{44}$$

q_1 drops out of every coefficient at the order that is asymptotically fixed.

2.6 Result

> *To the order of RCV's model, the depth-averaged system over a thin permeable rough bed is RCV (4.26)–(4.28) with*
>
> $$R \to R_p(d^+), \qquad \alpha \to \alpha_p = R_{1p} - R_p + 1, \qquad \alpha_1 \to \alpha_p - \alpha_2, \qquad \alpha_2 \text{ unchanged}, \tag{45}$$
>
> *and C_f given by (40). The 3-D reconstruction (RCV 5.3) keeps its form in the outer layer; in the inner layer and inside the bed the profile is $u_* F(z^+; d^+)$ plus the order-1 response.*

Each of the next three parts is therefore "solve one inner problem".

3 Step 2: null Reynolds stress in the bed

3.1 Sharp interface: closed form

Take a plane interface at $z = 0$, a homogeneous bed below with $\tau_t = \tau_d = 0$, and RCV's damped mixing length measured from the interface above. Below, Brinkman–Darcy for the superficial velocity u_D with effective viscosity $\nu_e = \beta_B \nu$ ($\beta_B = 1$ Neale–Nader, $\beta_B = 1/n_b$ Ochoa-Tapia–Whitaker):

$$\beta_B\, \nu\, u_D'' = \frac{\nu}{K} u_D \;\Rightarrow\; u_D = u_s\, e^{z/\delta_B}, \qquad \delta_B = \sqrt{\beta_B K}. \tag{46}$$

Continuity of velocity and of shear stress at $z = 0$ gives $\rho\beta_B \nu u_s/\delta_B = \tau_b$, i.e. a slip velocity

$$u_s^+ = \frac{Re_K}{\sqrt{\beta_B}}. \tag{47}$$

Above the interface the inner problem is RCV's with u^+ shifted by u_s^+ (the Van Driest layer is assumed to sit on the interface as on a wall). Therefore, exactly:

$$\boxed{R_p = R + \frac{\kappa\, Re_K}{\sqrt{\beta_B}}, \qquad \gamma = \frac{\kappa\, Re_K}{\sqrt{\beta_B}}, \qquad R_{1p} = \begin{cases} R_1 + 2\kappa Re_K/\sqrt{\beta_B} & \text{frozen damping} \\ R + 2\kappa Re_K/\sqrt{\beta_B} & \text{consistent} \end{cases}} \tag{48}$$

(the slip responds linearly to the wall stress, $\delta u_s^+/\delta S = Re_K/\sqrt{\beta_B}$, hence the factor 2 in R_{1p}). So $\alpha_p = \alpha + \kappa Re_K/\sqrt{\beta_B}$ (frozen) or $1 + \kappa Re_K/\sqrt{\beta_B}$ (consistent), and the friction law becomes

$$\frac{1}{\sqrt{C_f}} = \frac{1}{\sqrt{C_{f,\mathrm{smooth}}}} + \frac{Re_K}{\sqrt{\beta_B}}, \qquad Re_K = \frac{\sqrt{K\, gh \sin\theta}}{\nu}. \tag{49}$$

Interpretation. With no turbulence in the bed, permeability only produces slip: the flow is faster and *less* resistive than over a smooth wall. This is the macroscopic Beavers–Joseph picture and it is the right answer only for $Re_K \ll 1$. The order-0 profile inside the bed is $u_s e^{z/\delta_B}$, on top of RCV's inner profile above.

3.2 Diffuse interface (DANS)

With the full porosity profile (quintic smooth step from n_b at $z = -d_p$ to 1 at the crest $z = 0$), $\ell_t = 0$ wherever $n < 1$ and RCV's damped mixing length above the crest, (36) is solved by shooting from the bulk bed (decaying Brinkman mode) up to $z^+ = 2 \times 10^5$, adjusting the amplitude so that $S = 1$. Figure 1(a) and table 1 show that R_p grows quickly with d^+: the whole roughness layer is laminar and nearly drag-free near the crest, where $(1-n)^2$ is small. The null-Reynolds-stress model is therefore only usable for $d^+ \lesssim 10$, i.e. $Re_K \lesssim 0.3$; beyond that turbulence must penetrate, which is Step 3.

4 Step 3: Li & Sawamoto mixing length

Following RA (2.11)–(2.12),

$$\ell_t = \kappa Z_{LS}\Big[1 - \exp\Big(-\frac{\sqrt{Z_{LS}U/\nu}}{A^*}\Big)\Big], \qquad Z_{LS}(z) = \int_{-\infty}^{z} \frac{n - n_b}{1 - n_b}\,\mathrm{d}z, \qquad A^* = 26. \tag{50}$$

Three properties matter for the asymptotics.

1. Above the roughness layer $Z_{LS} = z - z_d$ with z_d inside the roughness layer (for my porosity profile, $z_d = -d_p/2$). The log law is in Z_{LS}, and by point (iii) of §2 the choice of origin is immaterial in the overlap.
2. Turbulence switches on continuously through the roughness layer; $\ell_t \to 0$ in the bulk bed, where the decaying Brinkman mode is still the deep-bed solution.
3. The damping depends on the local velocity, not on u_*. So there is no ambiguity in linearising: the consistent value $R_{1p} = R_p + \gamma$ is the natural one. Near a smooth wall $U^+ \simeq Z^+$, and the damping reduces to Van Driest's.

The inner problem is (36) with $\nu_d = 0$. Results are in figure 1 and table 1:

- $d^+ \to 0$: $R_p \to 3.46$ ($B \simeq 7.2$), not RCV's 2.67, because $\sqrt{Z^+U^+} < Z^+$ in the buffer layer damps more. For continuity with RCV one would recalibrate A^* (not done here).
- $d^+ \gtrsim 10^4$: fully rough, $R_p \simeq 3.87 - \ln d^+$ and $\gamma \to -1$. In the form $u^+ = \kappa^{-1}\ln(z/d_p) + B_d$, $B_d \to 8.2$; against Nikuradse's 8.5 this is an equivalent sand roughness $k_s \simeq 1.1\, d_p$.
- The relaxation coefficient $\alpha_p = 1 + \gamma$ goes from 1 (smooth) through a minimum $\simeq -0.65$ near $d^+ \approx 100$ back to 0 (fully rough). With frozen damping it goes from 3.3 to 0. Both agree in the fully rough limit, where damping is irrelevant. There $\alpha_1 = -\alpha_2$, so the relaxation terms of (28) keep only their α_2 (outer-layer) contributions.

5 Step 4: adding the dispersive stress

With RA's closure (2.13)–(2.15), $\nu_d^+ = \lambda_d\, d^+\,(1-n)/(1-n_b)\,|u^+|$ is added to the molecular viscosity in (36). It is zero above the crest, so the outer layer and the matching are untouched. Three remarks:

- ν_d depends on U, not on $\partial_z U$. The order-1 inner problem therefore picks up a term $\propto \delta U\, \partial_z U_0$, but the vector closure $\nu_d \propto |\boldsymbol{U}|$ is isotropic, so the parallel/normal split still holds.

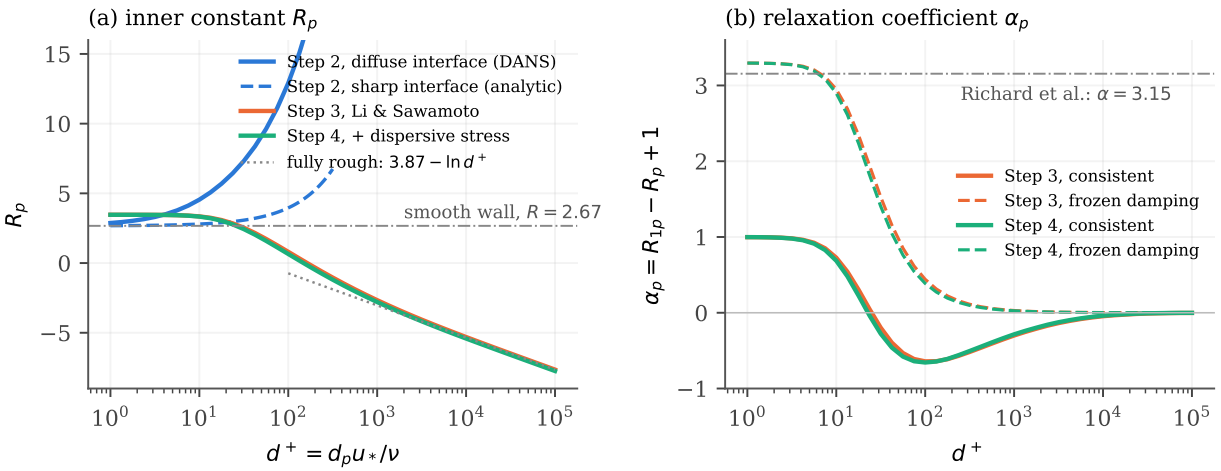


Figure 1: (a) Inner constant R_p versus d^+. Step 2: diffuse interface (DANS, $\ell_t = 0$ for $n < 1$) and sharp-interface closed form ($\beta_B = 1$). Steps 3–4: Li & Sawamoto mixing length without and with dispersive stress. Dotted line: fully rough asymptote. (b) $\alpha_p = R_{1p} - R_p + 1$ for steps 3–4, consistent linearisation (solid) and frozen damping (dashed). $n_b = 0.4$, $\kappa = 0.41$, $A^+ = A^* = 26$, $\lambda_d = 0.015$, roughness layer of thickness d_p.

Table 1: Inner constants. $B_p = (R_p - 1 + 2\ln 2 + \ln\kappa)/\kappa$; $\alpha_p^c = 1 + \gamma$ (consistent), α_p^f frozen damping. Smooth wall (RCV): $R = 2.669$, $B = 5.28$, $\alpha = 3.155$ (frozen), 1 (consistent).

Model	d^+	R_p	B_p	γ	α_p^c	α_p^f
Step 2 (DANS)	1	2.872	5.77	0.23	1.23	3.36
	10	4.547	9.86	1.64	2.64	4.80
	100	12.91	30.3	6.89	7.89	10.0
Step 3 (L&S)	1	3.463	7.21	0.00	1.00	3.30
	10	3.349	6.94	−0.27	0.73	2.95
	100	0.806	0.73	−1.64	−0.64	0.44
	10^3	−2.662	−7.72	−1.30	−0.30	0.03
	10^4	−5.306	−14.2	−1.05	−0.05	0.00
Step 4 (+ disp.)	1	3.463	7.21	0.00	1.00	3.30
	10	3.331	6.89	−0.31	0.69	2.90
	100	0.650	0.35	−1.66	−0.66	0.39
	10^3	−2.797	−8.05	−1.28	−0.28	0.03
	10^4	−5.415	−14.4	−1.04	−0.04	0.00

- Dispersion enhances momentum transfer in the roughness layer where the mixing length is small, so the bed is slightly rougher: R_p drops by 0.1–0.15 for $d^+ \gtrsim 100$, i.e. $k_s \simeq 1.25\, d_p$ instead of $1.1\, d_p$. γ and α_p are almost unchanged.
- RA's alternative (their eq. 5.1), a combined length $\ell_{t,D} = (\ell_{t,\mu}^2 + \ell_d^2)^{1/2}$ inside Prandtl's formula, makes the dispersive contribution $\propto |\partial_z U|\partial_z U$ instead of $U\partial_z U$. It changes only the inner ODE, not the framework.

With $\lambda_d = 0.015$ the effect on the depth-averaged model is small *in the large-submergence limit*. In RA's shallow-submergence experiments it was much larger because the roughness layer occupied a finite fraction of the depth (see the validity box in §2).

6 Validation against a full-depth solver

The asymptotic results were checked against an independent full-depth model: the steady 1-D DANS momentum balance $0 = n\rho g i + n\bar{f} + (\tau_v + \tau_t + \tau_d)' - \rho\nu n' u'$ on $[z_0, z_1]$. It uses $u = U_{\mathrm{SSL}}$ at the bottom and a stress-free surface, a tanh porosity profile ($n_b = 0.38$, transition over one d_p), the Li–Sawamoto depth $Z = \mathrm{softplus}(az)/a$, velocity-based damping, Ergun drag in the $\mu n/K$ form and RA's dispersive stress. The closure functions were taken unchanged from the supplied FEniCSx script. The weak form

was solved with P1 elements and Newton's method in NumPy, because FEniCSx is not available here. Its mixing length has no free-surface factor, so the outer profile is $u = (u_*/\kappa)[\ln Z^+ + f(s)] + u_* B_p$ with $f(s) = 2\sqrt{1-s} - 2\ln(1+\sqrt{1-s}) - 2 + 2\ln 2$. The inner problem (36) was solved with the same closures, and B_p was compared with $u/u_* - \kappa^{-1}[\ln Z^+ + f(s)]$ from the full model in the overlap. The predicted mean velocity is $U/u_* = \kappa^{-1}[\ln(hu_*/\nu) - 1 + \int_0^1 f\,\mathrm{d}s] + B_p$.

d^+	h/d_p	B_p (inner)	B_p (full model, overlap)	U/u_* (full)	U/u_* (asymptotic)
10	10^3	6.872	6.816 ($Z = 30d_p$)	26.183	26.213
10	10^4	6.871	6.871 ($Z = 100d_p$)	31.824	31.829
100	10^2	0.233	–	19.544	19.574
100	10^4	0.232	0.232 ($Z = 100d_p$)	30.804	30.806
1000	10^2	−8.000	–	16.958	16.956
1000	10^4	−8.001	−8.002 ($Z = 100d_p$)	28.187	28.188

Table 2: Inner constant and mean velocity: matched asymptotics versus full-depth solution.

h/d_p	3.75 (reference case)	10	30	100
error on U	+4.9%	+1.6%	+0.43%	+0.11%

Table 3: Decreasing submergence at fixed $d^+ \simeq 139$: relative error of the asymptotic mean velocity.

The inner constant is recovered to 3–4 digits once a genuine overlap exists, and the full solution stays within $0.2\%\,u_*$ of the composite asymptotic profile. Because $B_p(d^+)$ matches across d^+, its slope γ, and hence the consistent $\alpha_p = 1+\gamma$, are confirmed as well. The frozen-damping alternative and the order-1 terms concern unsteady, non-uniform flows and cannot be tested with a steady 1-D solver.

7 Summary

1. RCV's chain: outer log layer (the $(1-s)$ free-surface factor cancels the linear stress) → viscous inner layer with constant stress → matching at $s \sim \sqrt{\eta} \to \mu = 2/\ln(\kappa^2 Re)$ → order 1 as a linear response to the wall stress → depth averaging with an enstrophy tensor → elimination giving relaxation source terms. The constants are R (secant response) and R_1 (tangent response).
2. A thin permeable rough bed changes only the inner problem: the drag is $O(1)$ there, while gravity, unsteadiness and the subsurface flux are $O(\eta)$. The model keeps RCV's form with $R \to R_p(d^+)$, $R_1 \to R_{1p}(d^+)$ and an explicit friction law (40). The h-dependence of R_p modifies the outer T_1, T_2 through $q_1 = \frac{3}{2} + \frac{\gamma}{2}$, but not the coefficients at the order that is fixed.
3. Null Reynolds stress in the bed gives slip and *lower* resistance ($1/\sqrt{C_f}$ increased by $Re_K/\sqrt{\beta_B}$ for a sharp interface). It is only valid for $Re_K \ll 1$.
4. The Li & Sawamoto length gives a smooth-to-fully-rough transition with $k_s \simeq 1.1\,d_p$, $\gamma \to -1$ and $\alpha_p \to 0$. The dispersive stress adds a small extra resistance ($k_s \simeq 1.25\,d_p$).
5. A full-depth DANS solver reproduces B_p and the mean velocity to 10^{-3} at large submergence, and to 5% even at $h/d_p = 3.75$.
6. Open choice: frozen versus consistent linearisation of the damping changes α_p substantially for $d^+ \lesssim 10^3$.

Reproducibility. Symbolic checks (sympy): order-1 outer stress for general $Q(h)$; U_1, $\varphi^{(1)}$; elimination for c_1, c_2; dissipation-tensor elimination for e_1, e_2. Numerics (scipy, LSODA, shooting): smooth-wall check $R = 2.6689$ and $R_1 = 4.8237$; inner problems of Steps 2–4 for $1 \le d^+ \le 10^5$; the identity $R_{1p} = R_p + \gamma$ was checked against the linearised ODE.